

Multi-Page Visually Rich Document Understanding

A survey of system design and an empirical attribution of failure

Lewei Xu

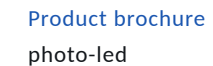
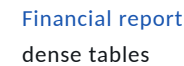
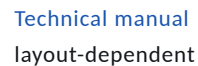
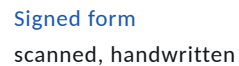
Supervised by Dr. Yihao Ding · Dr. Daochang Liu



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- Survey of MP-VRDU
- Empirical Attribution Study
- Proof of Concept: A Simple Method
- Conclusion

Meaning is carried jointly by **text, vision, and layout** rather than by text alone. **Multi-page** adds a further layer: meaning also runs between pages, as when a claim points to a figure many pages away.



- **Text.** Small, typeset or handwritten, and legible only at far higher input resolution than a typical vision-language task requires.
- **Vision.** Charts, figures, photographs, and handwriting carry meaning that no text extraction recovers.
- **Layout.** Position is semantic, and reading order must be recovered rather than assumed.

What is Document Understanding?

Recover **what is on each page** and **how the whole document is organised**, then answer questions over it.

LAYOUT ANALYSIS

3

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Latinos Increasingly Confident in Personal Finances, See Better Economic Times Ahead

Yet many economic indicators show few gains for the community since the Great Recession

Latinos have become considerably more upbeat about their personal finances and optimistic about their financial future since the Great Recession, according to newly released results from a national survey of Latino adults. The survey also shows that Latinos have pulled even with the general U.S. population in their views of their personal finances and continue to outpace them on optimism about the future. However, community economic indicators show limited progress since the Great Recession.

Four-in-ten Latinos say their personal finances are in "excellent" or "good" shape, a 17 percentage point increase since 2008, when only 23% made a similarly positive assessment of their finances (the Great Recession began in December 2007).¹ By contrast, the share of all Americans who have a similarly positive view of their finances remained essentially flat during the same seven-year period (41% in 2008 vs. 43% in 2015).

The survey also shows that Hispanics are more upbeat in their financial expectations for the upcoming year than they were in 2008. About eight-in-ten Hispanic adults (81%) say they expect their family's financial situation to improve "a lot" or "some" in the next year, up from 67% who said the same in 2008. By comparison, the U.S. public is not as upbeat –

Hispanics more confident in personal finances than in 2008, see better economic times ahead

% who say their personal financial situation is in excellent/good shape

% who expect their family's financial situation to improve "a lot" or "some" in the next year

General public

Hispanic

RECESSION

2004

2008

2011

2015

2004

2008

2011

2015

51%

41%

43%

43%

31%

23%

24%

33%

70%

67%

79%

81%

66%

54%

67%

61%

Note: Shaded area marks the Great Recession, which ran from December 2007 to June 2009.

Source: National Survey of Latinos conducted Oct. 23-Nov. 30, 2015; surveys conducted Dec. 8-13, 2010; Oct. 24-28, 2012; Sept. 12-16, 2013; Dec. 7-11, 2011; Dec. 3-8, 2008; Oct. 19-22, 2006; Aug. 5-10, 2004.

¹Latinos Increasingly Confident in Personal Finances, See Better Economic Times Ahead

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The Great Recession lasted from December 2007 to June 2009, according to economists at the National Bureau of Economic Research.

www.fewresearch.org

DOCUMENT OUTLINE

Research paper, 24 pages

5.2 Conscious Incompetence

Paragraphs

Figure 4

Caption

5.3 Retrieval Analysis

Paragraphs

Figure 5

Caption

5.4 Human Evaluation

Paragraphs

Table 6

Caption

Title Section heading Text Figure / chart Table Caption / note Header / footer

LLM Assistants

Provide a demonstration of how you would use tool calls to answer a difficult question over this long visually rich document.

Designing a sample tool-based walkthrough for a cross-page question.

I'll demonstrate on a question that needs a figure and a table: "**How often does the final pipeline stop at its earliest allowed point, and how does its page usage compare with the number of gold evidence pages?**" First I'll map printed page numbers to PDF pages, then read the relevant pages as images.

Read a file, updated tasks ▾

Added task **Locate pages for Figure 4.2 and Table 4.2** ▾

Request

Map printed page numbers to PDF pages using the contents list and the offset seen on page 27

Added task Read figure and table pages visually

Request

Description	Read PDF pages 60-61 and extract the bar heights and token statistics
-------------	---

Added task **Verify and answer** ▾

Request

Question	Answer	Description
1. The number of people who are not in the workforce is 10 million. The number of people who are in the workforce is 20 million. The number of people who are unemployed is 5 million. What is the unemployment rate?	25%	Cross-check figure values against surrounding text, then answer

Completed task Locate pages for Figure 4.2 and Table 4.2

Started task Read figure and table pages visually

Read Honours Lewei Xu.pdf >

Verifying the data points plotted across the chart.

Completed task Read figure and table pages visually

Completed task Verify and answer

- A question over a **100-page thesis**: the assistant plans, locates the pages, then answers.
- It **renders the pages and reads them as images**.

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[illegible]

Table 4.1 MMLongBench-Doc accuracy (%) for published systems using open 7-8B vision-language backbones. Reported values are taken directly from the cited papers; “vis” indicates results reproduced by another study rather than reported in the system’s original paper. **OCR**: parser or OCR text is provided as input. **Trained**: task-specific fine-tuning or RL within the pipeline. ^aA text LLM operates alongside the VLM. ^bRetriever only. ^cThe document outline includes visual-element descriptions generated by GPT-3.5-Turbo

and layout questions decline. Multi-page accuracy also remains a clear weakness at 37.8%, below Doc Trace's 41.1%, indicating that the higher overall accuracy does not eliminate the underlying difficulty of integrating evidence across pages.

Adaptive evidence acquisition. Figure 4.2 shows that the pipeline does not uniformly consume its full page budget. The largest gaps occur on the first six pages, the earliest point permitted by the stoppage rule, while generally fewer questions are asked deeper into the textual ranking and later in the reading process (e.g. 25–35). Revisited trajectories account for a part of this long tail, with some questions requiring one or multiple CONTINUE decisions after an initial synthesis attempt. This differs substantially from the annotated evidence distribution, where most answerable questions contain one or two gold pages. The gap reflects the practical setting in which the locations of those pages are unknown and additional contextual pages may also contribute useful evidence.

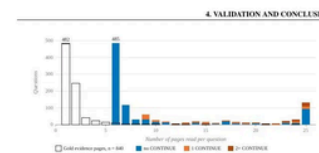


Figure 4.2 Trajectory of our method

	Tokens				Latency (s)		Peak VRAM (GiB)
	Turns	Input text	Input visual	Output	Prefill	Decode	
reader	11.2	5.209	834	1,808	2.0	70.6	7.3
synthesizer	1.4	9.638	1,342	724	4.0	59.0	8.1
arbiter	1.3	4.988	0	1,083	1.6	72.8	7.7
Per question, mean	13.9	78.215	11,733	13,721	29.7	968.1	8.4
Per question, median	0.0	37.404	7,737	8,038	13.8	544.6	8.1

Table 4.2 Inference cost of the pipeline per agent turn and question

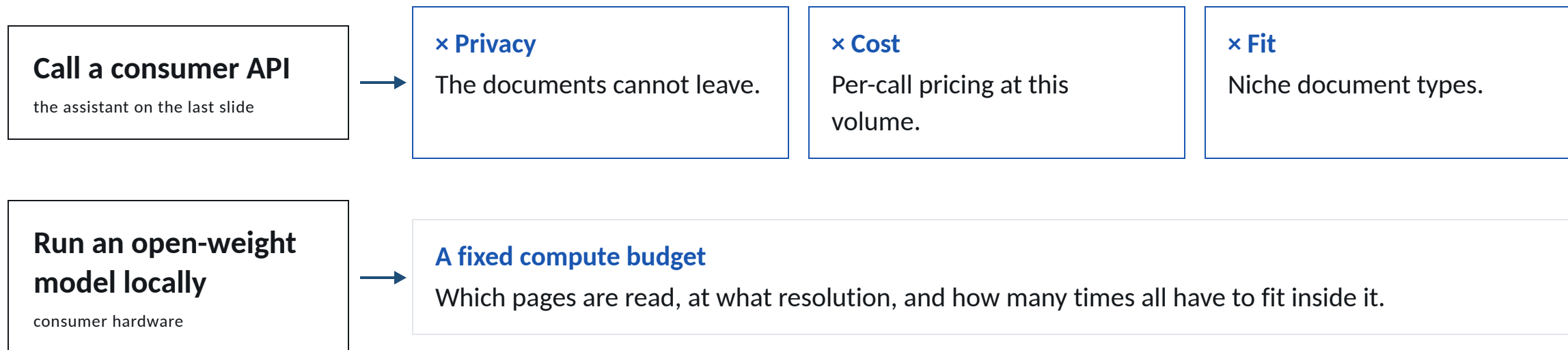
Inference cost. Table 4 quantifies the cost of this adaptive behavior. The pipeline uses 13.9 turns per question on average, compared with a median of 9.6, with the reader accounting for 11.2 turns and the synthesizer and arbiter involved only 1.4 and 1.3 times per question respectively. Mean peak memory is 8.2 GiB, remaining comfortably within the 16 GiB budget of the deployment GPU, while the average question costs 78.215 text-input tokens, 11.733 visual tokens, and 13.921 output tokens across its trajectory. The principal cost is latency rather than memory: mean prefill time is only 29.7 s per question compared with 48.1 s of queuing, with a median total of roughly 9.3 minutes. The method also trades additional sequential inference for broader memory use and broader evidence acquisition, with the deployment parameters providing a direct mechanism for reducing trajectory depth where lower latency is required.

4.2 Open Challenges and Future Work

The survey of Chapter 2 identifies five challenges that define the open MP-VDRU frontier. They arise from a consistent pattern across the surveyed systems: progress comes from moving evidence handling out of the backbone, through compression, retrieval, or iterative orchestration, but each shift relocates the bottleneck rather than removing it. Chapter 3 and the system-level validation in §4.1 address parts of the first three challenges directly, but also show that their underlying trade-offs remain unresolved. Three concern system capability, in the evidence pipeline itself, in faithfulness, and in efficiency. Two concern data and evaluation, in supervision integrity and in evaluation scope.

Real-World Constraints

A hospital, a mining company, a law firm: **tens of thousands of documents** to classify, summarise, or extract from.



The research question: document understanding **under a fixed compute budget**.

Table of Contents

- Introduction

- **Survey of MP-VRDU**

Managing Evidence at Document Scale: A Survey of Multi-Page Visually Rich Document Understanding.

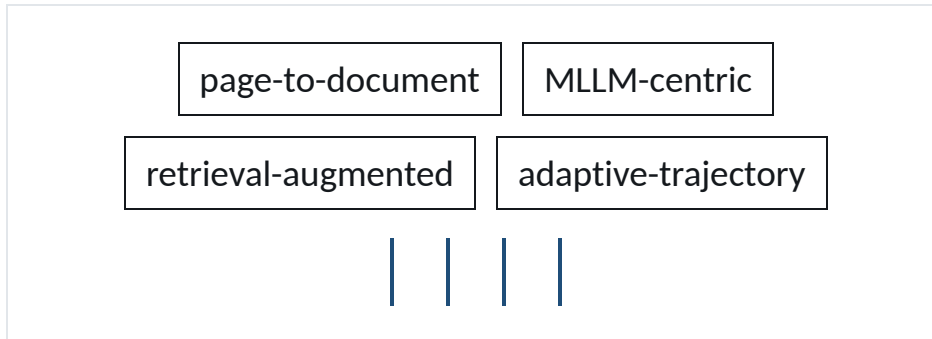
Lewei Xu, Yihao Ding, Zihan Xu, Siwen Luo, Daochang Liu, Wei Liu

[EMNLP 2026, main conference](#)

- Empirical Attribution Study
- Proof of Concept: A Simple Method
- Conclusion

Survey: Research Gap and Contributions

RESEARCH GAP



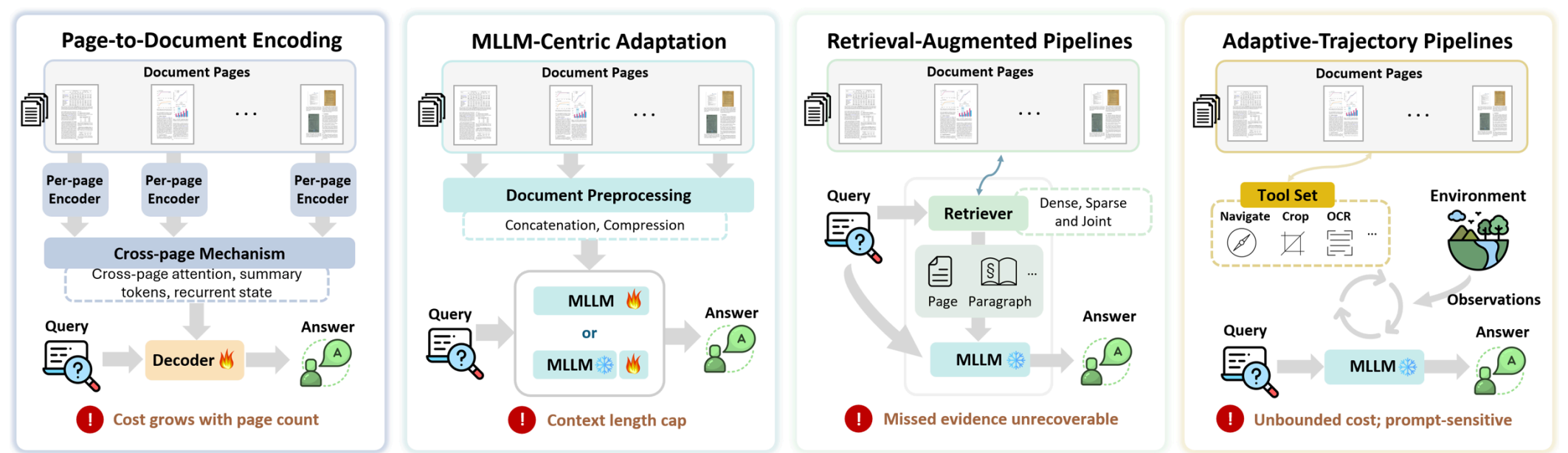
Multi-page systems grew along **parallel lines**, with inconsistent terminology. No survey treats multi-page as its own problem.

CONTRIBUTIONS

- 1 Defines multi-page understanding as a problem of **evidence management**.
- 2 A taxonomy: four architectural families, multimodal representation, inference-time and training strategies.
- 3 Consolidates the datasets and five open challenges.

Multi-Page Architectural Overview

The survey organises systems by **evidence management**, into four architectural families. **Page-to-document encoding models** jointly encode pages; **MLLM-centric adaptation models** fit multi-page input into an MLLM’s context and token budget; **retrieval-augmented pipelines** retrieve evidence before generation; **adaptive-trajectory pipelines** inspect evidence iteratively under an LLM controller.



Snowflake indicates typically frozen, fire indicates typically trained

Surveyed Systems at a Glance

Model	Venue	OCR	Vision Encoder	LLM Backbone	Mod.	Search	Reasoning	Agentic
Page-to-Document Encoding Models								
Arctic-TILT (2025)	ACL	Yes	U-Net	T5-Large	T, V, L	-	-	-
GRAM (2024)	CVPR	Yes	DocFormerv2	DocFormerv2	T, V, L	-	-	-
Hi-VTS (2023)	PR	Yes	DiT	T5-Base	T, V, L	-	-	-
RM-T5 (2024)	DAS	Yes	DiT	T5-Base	T, V, L	-	-	-
MLLM-Centric Adaptation Models								
DocR1 (2026)	AAAI	No	Qwen2.5-VL	Qwen2.5-VL-7B	V	-	CoT, HD	-
DocSeeker (2025)	CVPR	No	Qwen2.5-VL ViT	Qwen2.5-VL-7B-Instruct	V	-	CoT, HD	-
Leopard (2024)	TMLR	No	SigLIP-SO400M	LLaMa-3.1-8B, Mistral-7B	V	-	CoT	-
mPLUG-DocOwl2 (2025)	ACL	No	ViT-L/14	mPLUG-Owl2	V	-	-	-
URaG (2026)	AAAI	No	Qwen2.5-VL ViT	Qwen2.5-VL-3B / Qwen2.5-VL-7B	V	-	HD	-
Docopilot (2025b)	CVPR	No*	InternViT-300M	Multiple (Intern)	V	-	-	-
Texthawk2 (2024)	preprint	No*	SigLIP-SO400M	Qwen2.5-VL-7B-Instruct	T, V, L	-	-	-
DocSLM (2025)	CVPR	Yes	SigLIP2	Qwen2.5-1.5B	T, V, L	-	-	-
InstructDoc (2024)	AAAI	Yes	CLIP	FlanT5, BLIP2	T, V, L	-	-	-
LayTokenLLM (2025b)	CVPR	Yes	-	LLaMA3-8B, Qwen1.5-7B	T, L	-	-	-
CoR (2026)	preprint	Yes*	Qwen2.5-VL	Qwen2.5-VL-7B	T, V	-	CoT, PE, HD	-
Retrieval-Augmented Pipelines								
AVIR (2025)	MMAAsia	No	Pix2Struct	Qwen2.5-VL-3B-AWQ	V	D	-	-
DFVC (2026)	Electronics	No	VisRAG, ColQwen	Qwen2-VL-2B-Instruct	V	D	-	-
DREAM (2025)	MM	No	InternVL2 ViT	InternVL2-4B	V	J	HD	-
M3DocRAG (2024)	preprint	No	ColPali, ColQwen	Multiple (Qwen/Intern/Idefics)	V	D	-	-
MM-R5 (2026)	AAAI	No	Qwen2.5-VL ViT	Qwen2.5-VL-7B	V	D	CoT	-
MoLoRAG (2025b)	EMNLP	No	ColPali	Multiple (Qwen/DS/LLaVA)	V	D, Nav	-	1A
Self-Attention Scoring (2024)	ICDAR	No	Pix2Struct	Pix2Struct	V	D	-	-
SLEUTH (2025)	preprint	No	ColPali, Qwen3-VL ViT	Multiple (Qwen/GLM/Gemini)	V	D	HD	4A
SV-RAG (2025)	ICLR	No	InternVL2, Phi-3-vision	Multiple (Intern/Pali/Phi)	V	D	-	-
VDocRAG (2025)	CVPR	No*	Phi3V-4.2B	Phi3V-4.2B	V	D	-	-
CREAM (2024)	MM	Yes	Pix2Struct	LLaMA2-7B	T, V	J	HD	-
KGP (2024c)	AAAI	Yes	-	Multiple	T, L	S, IR, Nav	-	1A
MHier-RAG (2025)	preprint	Yes	Qwen-VL-Plus	Multiple (Qwen/DS/GPT)	T, V, L	J	CoT, HD	-
MLDocRAG (2026)	preprint	Yes	Qwen2.5-VL	Multiple (Qwen)	T, V, L	D	CoT, HD	-
MultiDocFusion (2025)	ACL	Yes	-	Multiple (Qwen/LLaMa/Mistral)	T, V, L	D	HD	-
M2RAG (2025a)	NeurIPS	Yes*	VisRAG-Ret	Multiple (Qwen)	T, V	J	-	3A
MDocAgent (2025)	preprint	Yes*	ColPali, ColQwen2	Multiple (Llama/Qwen)	T, V	J	-	5A
PDF-WuKong (2024)	preprint	Yes*	IXC2-VL-4KHD	IXC2-VL-4KHD	T, V	J	-	-
RAG-DocVQA (2025)	ICDAR	Yes*	DiT, Pix2Struct	Qwen2.5-VL-7B-Instruct	T, V, L	D	-	-
VisDoMRAG (2025)	NAACL	Yes*	ColPali / ColQwen2	Multiple (Qwen/Gemini/GPT)	T, V	J	CoT	3A, SC
Adaptive-Trajectory Pipelines								
Doc-V* (2026)	preprint	No	Qwen2.5-VL	Qwen2.5-VL-7B-Instruct	V	D, IR, QR, Nav	ReAct, HD	1A
DocReact (2025a)	ACL	No	ColPali, VisRAG	GPT-4o	V	D, IR, QR	ReAct, QD	1A
MACT (2025)	preprint	No	Multiple	Multiple (Qwen/MiMo/Intern)	V	D	CoT, PE	4A, SC, SV
VRAG-RL (2026)	NeurIPS	No	Qwen2.5-VL ViT	Qwen2.5-VL-3B/7B-Instruct	V	D, IR, QR	ReAct, HD	1A
DocLens (2025a)	preprint	Yes	-	Multiple (GPT/Gemini/Claude)	T, V, L	J	CoT	4A, SA
DMAP (2026)	WWW	Yes*	ColPali	GPT-4o / Qwen-plus	T, V, L	J	HD	2A, SR
DocAgent (2025)	EMNLP	Yes*	-	Multiple (GPT/Gemini/Claude)	T, V, L	J, IR, Nav	ReAct, HD	3A
DocDancer (2026)	preprint	Yes*	Qwen3-VL-235B	Qwen3-4B/30B-A3B	T, V, L	J, IR, QR	ReAct	1A
MM-Doc-R1 (2026)	preprint	Yes*	Qwen2.5-VL ViT	Qwen3-4B/8B	T, V	S, IR, QR, Nav	ReAct, HD	3A
SimpleDoc (2025)	EMNLP	Yes*	ColQwen-2.5	Multiple (Qwen)	T, V	J, IR, QR	-	1A, SR, SV
ViDoRAG (2025a)	EMNLP	Yes*	NV-Embed-V2, ColQwen2	Multiple (Qwen/LLaMa/GPT)	T, V	J, IR	ReAct, HD	3A, SR, SV

1 Modalities

Encoding models fuse **text, vision and layout**. MLLM-centric models are mostly **vision only**.

2 Search, fixed

Retrieval-augmented pipelines add one **dense or joint** retrieval step.

3 Search, adaptive

Adaptive-trajectory pipelines add **iterative retrieval, query reformulation and navigation**.

4 Reasoning

ReAct and **multi-agent** designs concentrate in the last family.

Page-to-Document Encoding: Hi-VT5

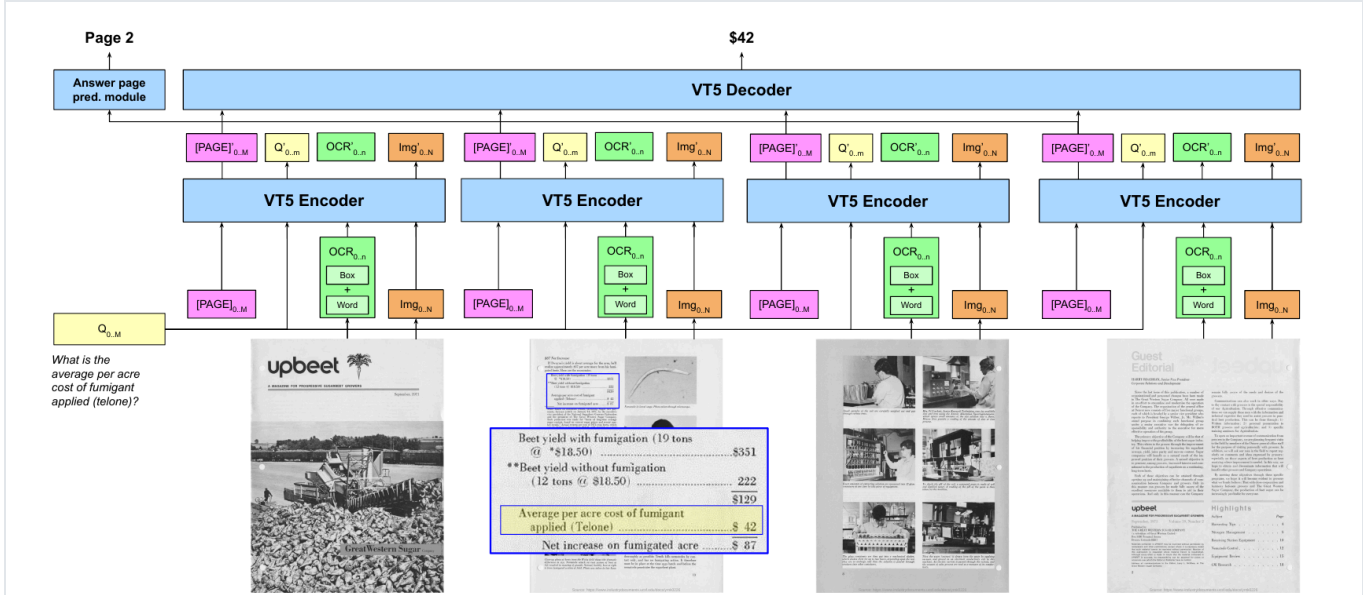
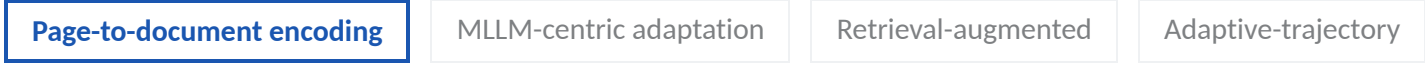


Figure from Tito et al., Pattern Recognition 2023

HOW IT WORKS

- Each page is encoded on its own from OCR text, layout boxes and image patches.
- Learnable [PAGE] tokens summarise each page. The decoder reads only those.

LIMIT

A page is squeezed into a few tokens, and **cost grows with page count**.

MLLM-Centric Adaptation: Docopilot

Page-to-document encoding

MLLM-centric adaptation

Retrieval-augmented

Adaptive-trajectory

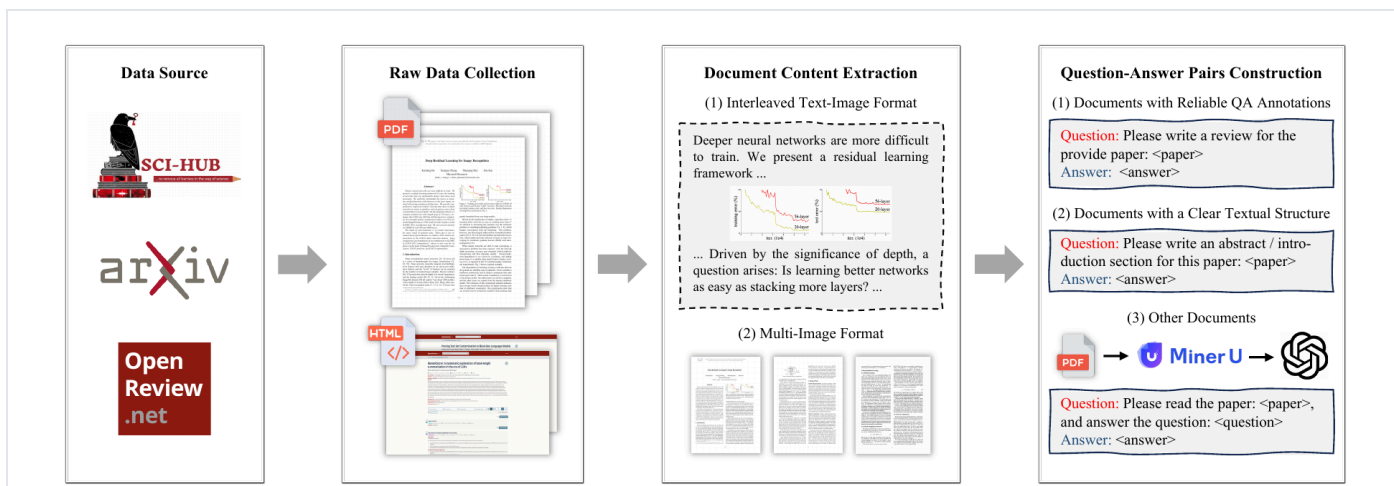


Figure from Duan et al., CVPR 2025

HOW IT WORKS

- A **native multi-page MLLM**: the whole document goes into the context, with no retrieval step.
- Trained on **Doc-750K**, built from papers on Sci-Hub, arXiv and OpenReview.

LIMIT

Bounded by the **context window**, and accuracy drops well before it is full.

Retrieval-Augmented: MoLoRAG

Page-to-document encoding

MLLM-centric adaptation

Retrieval-augmented

Adaptive-trajectory

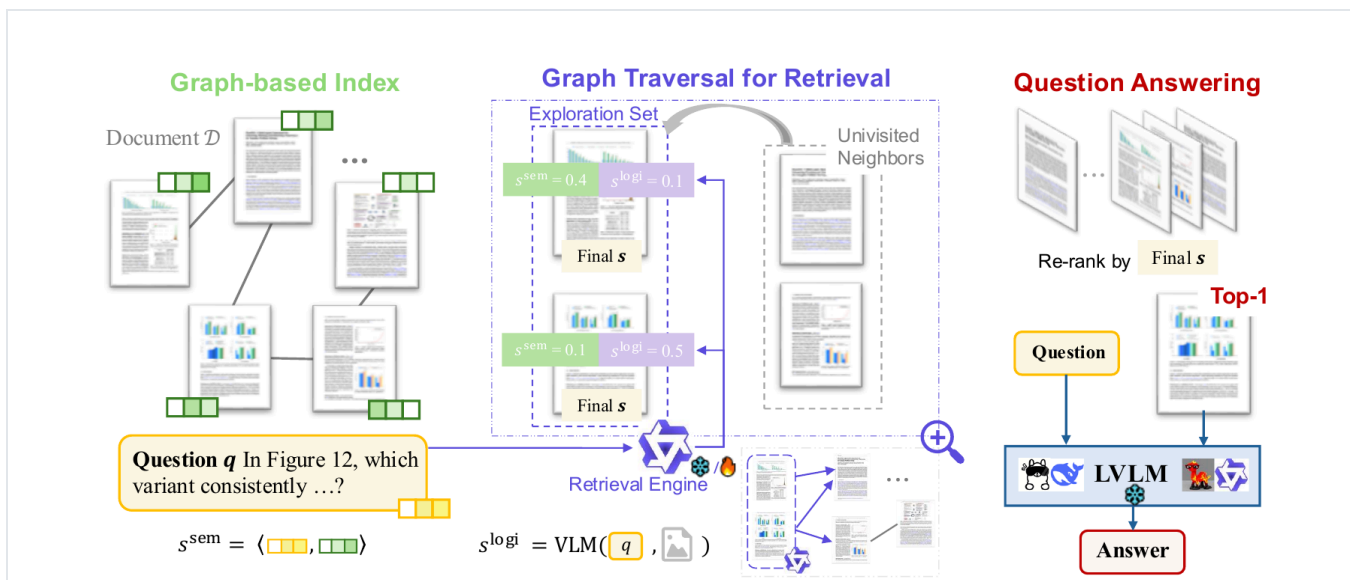


Figure from Wu et al., EMNLP 2025

HOW IT WORKS

- Pages form a **graph**. A VLM walks it from the most similar pages out to their neighbours.
- Each page gets a **semantic and a logical** relevance score. The top pages go to a frozen model.

LIMIT

Evidence the traversal misses is **unrecoverable** at generation.

Adaptive-Trajectory: DocLens

Page-to-document encoding

MLLM-centric adaptation

Retrieval-augmented

Adaptive-trajectory

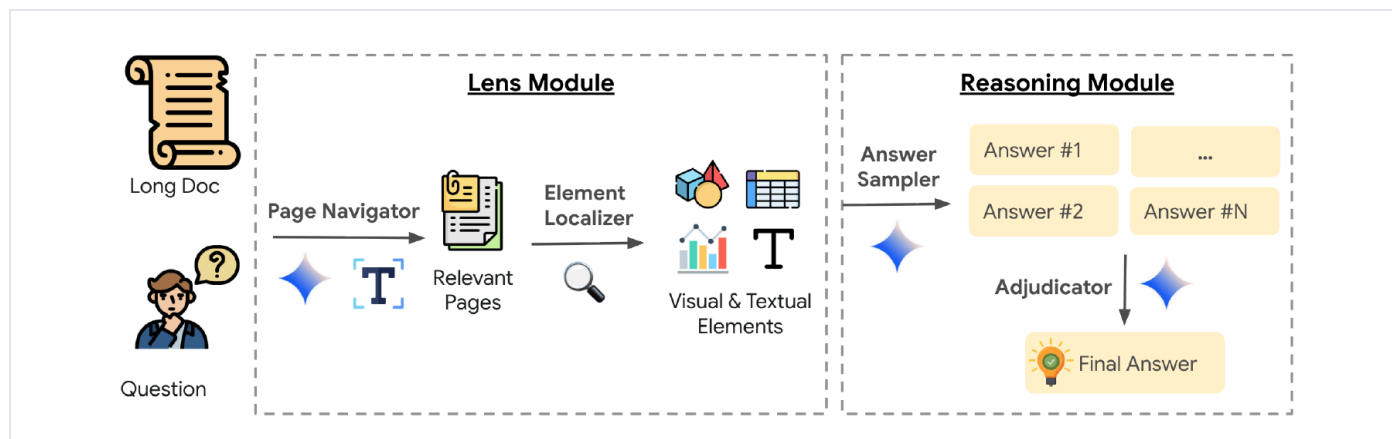


Figure from Zhu et al., ACL 2026

HOW IT WORKS

- **Lens module**: a page navigator finds the pages, then an element localiser zooms into figures, tables and text.
- **Reasoning module**: several answers are sampled and an adjudicator picks one.

LIMIT

Many model calls per question, so **cost is hard to bound**.

Table of Contents

- Introduction
- Survey of MP-VRDU

- **Empirical Attribution Study**

Locating Failure in Multi-Page Visually Rich Document Understanding: An Empirical Attribution

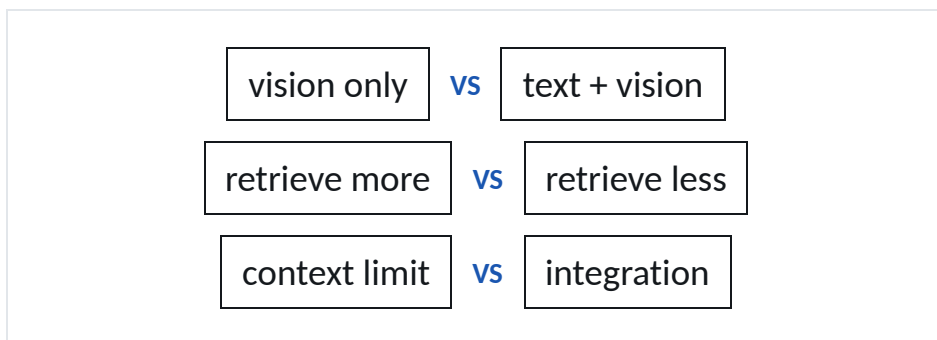
Lewei Xu, Yihao Ding, Zihan Xu, Daniel Yitian Su, Daochang Liu, Siwen Luo, Yifan Peng, Wei Liu

[Submitted to EACL, under review](#)

- Proof of Concept: A Simple Method
- Conclusion

Empirical Study: Research Gap and Contributions

RESEARCH GAP



Papers make **competing claims** about what matters, each on a different pipeline and dataset, so none can be compared.

CONTRIBUTIONS

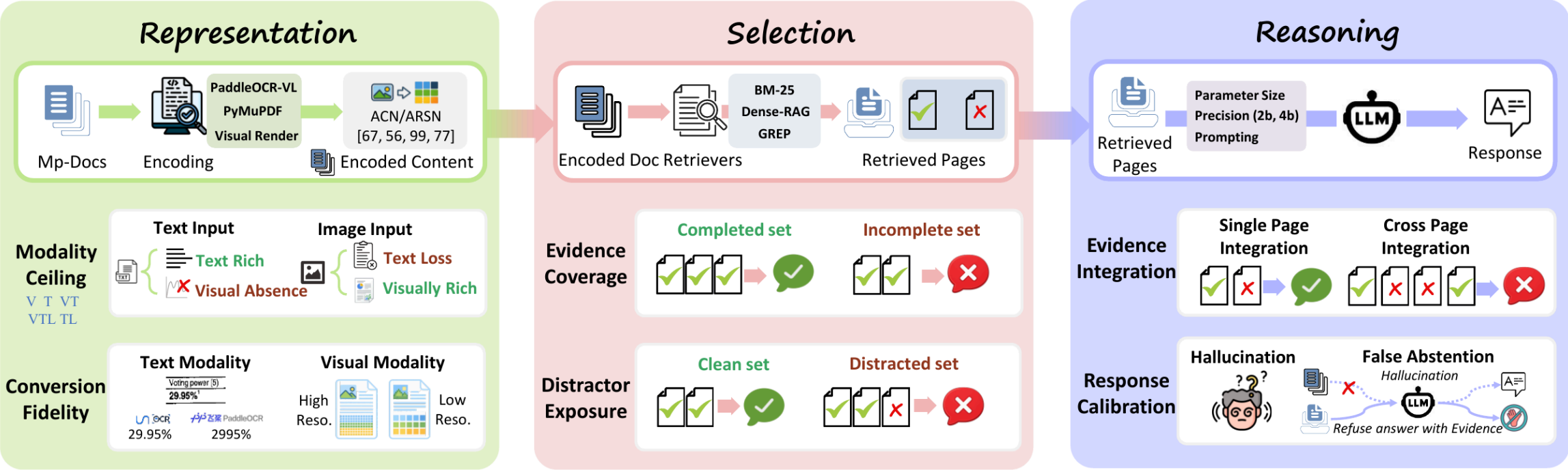
- 1 Three architecture-independent **failure loci**, with six mechanisms.
- 2 Controlled interventions in one pipeline: change one locus, hold the others fixed.
- 3 Transfer across datasets and models, with design implications and cost.

	topics, tests, essays, building on other ideas and expressing their own clarity and confidence. 5. Reading and Learning 11.17.2: Integrate multiple sources of information presented in diverse formats and media (e.g., visually, quantitatively, orally) in order to make informed decisions and solve problems, evaluating the credibility and accuracy of each source and noting any discrepancies among the data. Reading for Informational Text 11.17.2: Analyze a complex or multiple of texts of sequence or selection and explain how those individual texts, ideas, or events interact and develop over the course of time. 6. Reading for Informational Text 11.17.2: Determine the meaning of words and phrases as they are used in the text, including technical meanings, analyze how an author uses and refines the meaning of a word or terms over the course of a text. 7. Reading for Informational Text 11.17.2: Analyze the main ideas and supporting details presented in diverse media and formats (e.g., visually, quantitatively) as well as words in order to address a question or solve a problem.	• Research them
SUMMARY OF ALL KEY ASSIGNMENTS		
QUARTER 1	QUARTER 2	QUARTER 3
UNIT 1: Predisposing Your Source: TED Talk, The: The Power of Fusion and Neuroscience, said a talker, refers to what you do in Google Classroom. UNIT 1&2 Team Activity: Form 5 to seven teams. Each team is responsible for each of the sections of the external Business Environment discussed in the chapter (economics, politics, culture, demographics, social structure, technology, and the environment). (Interpersonal, Resources, Information).	Unit 2: Preparing Your Work: Business Plans at the Google Classroom. Unit 3: Preparing Your Work: Business Plans at the Google Classroom. Unit 4: Preparing Your Work: Business Plans at the Google Classroom. Unit 5: Preparing Your Work: Business Plans at the Google Classroom. Unit 6: Preparing Your Work: Business Plans at the Google Classroom. Unit 7: Preparing Your Work: Business Plans at the Google Classroom. Unit 8: Preparing Your Work: Business Plans at the Google Classroom. Unit 9: Preparing Your Work: Business Plans at the Google Classroom. 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UNIT 3: Preparing Your Work: Business Plans at the Google Classroom. UNIT 4: Preparing Your Work: Business Plans at the Google Classroom. UNIT 5: Preparing Your Work: Business Plans at the Google Classroom. UNIT 6: Preparing Your Work: Business Plans at the Google Classroom. UNIT 7: Preparing Your Work: Business Plans at the Google Classroom. UNIT 8: Preparing Your Work: Business Plans at the Google Classroom. UNIT 9: Preparing Your Work: Business Plans at the Google Classroom. UNIT 10: Preparing Your Work: Business Plans at the Google Classroom. UNIT 11: Preparing Your Work: Business Plans at the Google Classroom. UNIT 12: Preparing Your Work: Business Plans at the Google Classroom. UNIT 13: Preparing Your Work: Business Plans at the Google Classroom. UNIT 14: Preparing Your Work: Business Plans at the Google Classroom. UNIT 15: Preparing Your Work: Business Plans at the Google Classroom. UNIT 16: Preparing Your Work: Business Plans at the Google Classroom. 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Correct answer: 6. Suppose the system says 4.

End-to-end accuracy **cannot tell these apart**, and each needs a different fix.

Attribution Framework



REPRESENTATION

Was the information preserved when the page was converted?

SELECTION

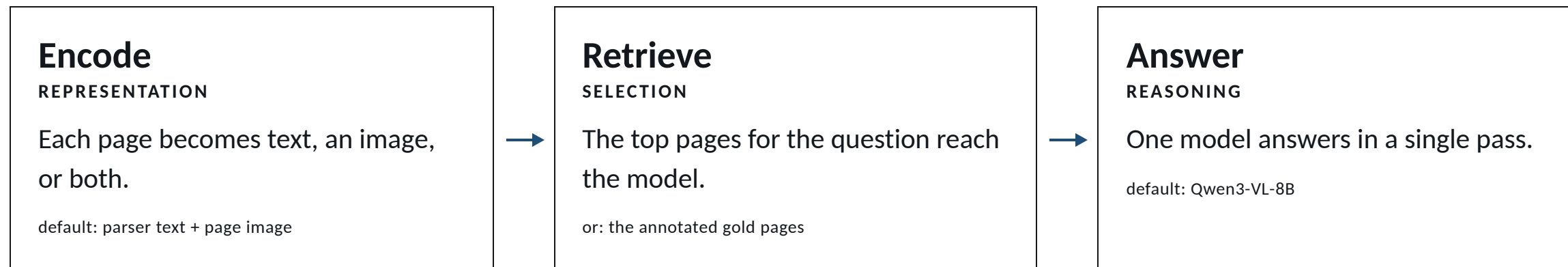
Did it actually reach the model?

REASONING

Did the model use it correctly?

Attribution by Construction

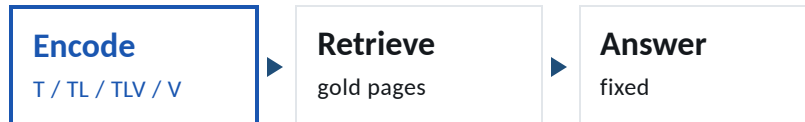
One simple pipeline. **Change one stage, hold the other two fixed.**



- **Gold pages** bypass retrieval, so selection can be taken out of the picture.
- **MMLongBench-Doc**: 135 documents, annotated with evidence pages, evidence type and answerability.

Modality Ceiling

Evidence	Abstention				Accuracy			
	T	TL	TLV	V	T	TL	TLV	V
Text	23.0	11.5	7.3	8.0	36.9	44.4	51.0	42.3
Layout	20.3	10.2	0.8	4.2	26.3	33.9	49.2	43.2
Table	15.9	7.2	6.7	9.1	35.6	44.7	44.7	34.1
Chart	32.6	28.1	7.3	6.2	16.3	16.9	42.7	40.4
Figure	36.6	24.8	7.6	7.9	14.5	22.1	42.4	39.0
<i>All</i>	26.3	17.2	6.2	7.1	29.0	35.7	49.5	42.6



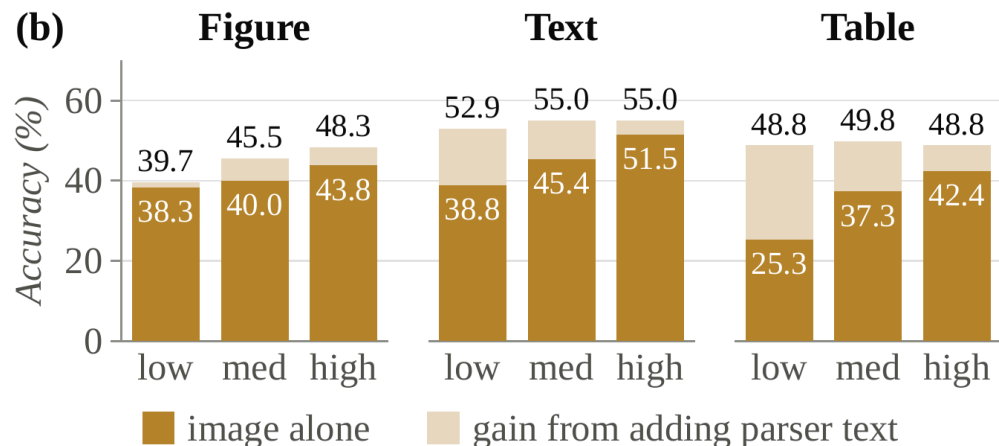
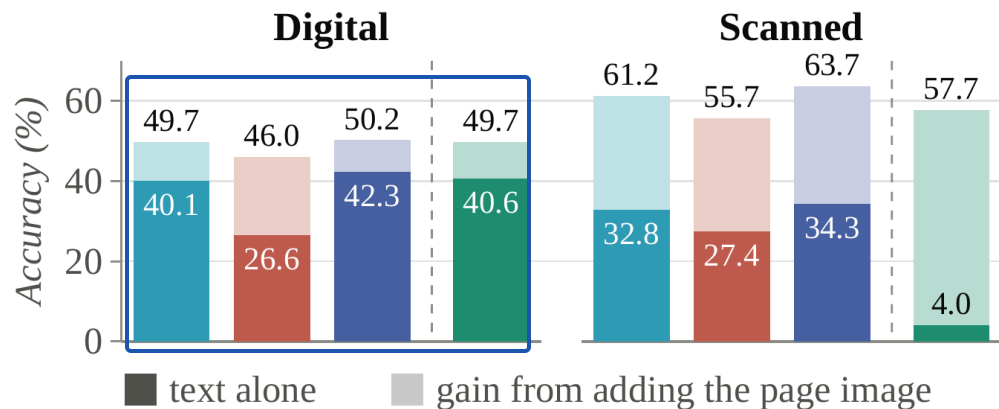
REPRESENTATION

Evidence accessibility depends on modality

- Text-only input refuses **chart and figure** questions most. Adding vision brings refusal **under 10%**.
- Images alone stay below text plus images: **42.6% against 49.5%**.

Conversion Fidelity

(a) PaddleOCR MinerU Unlimited-OCR PyMuPDF



Encode
parser, resolution

Retrieve
gold pages

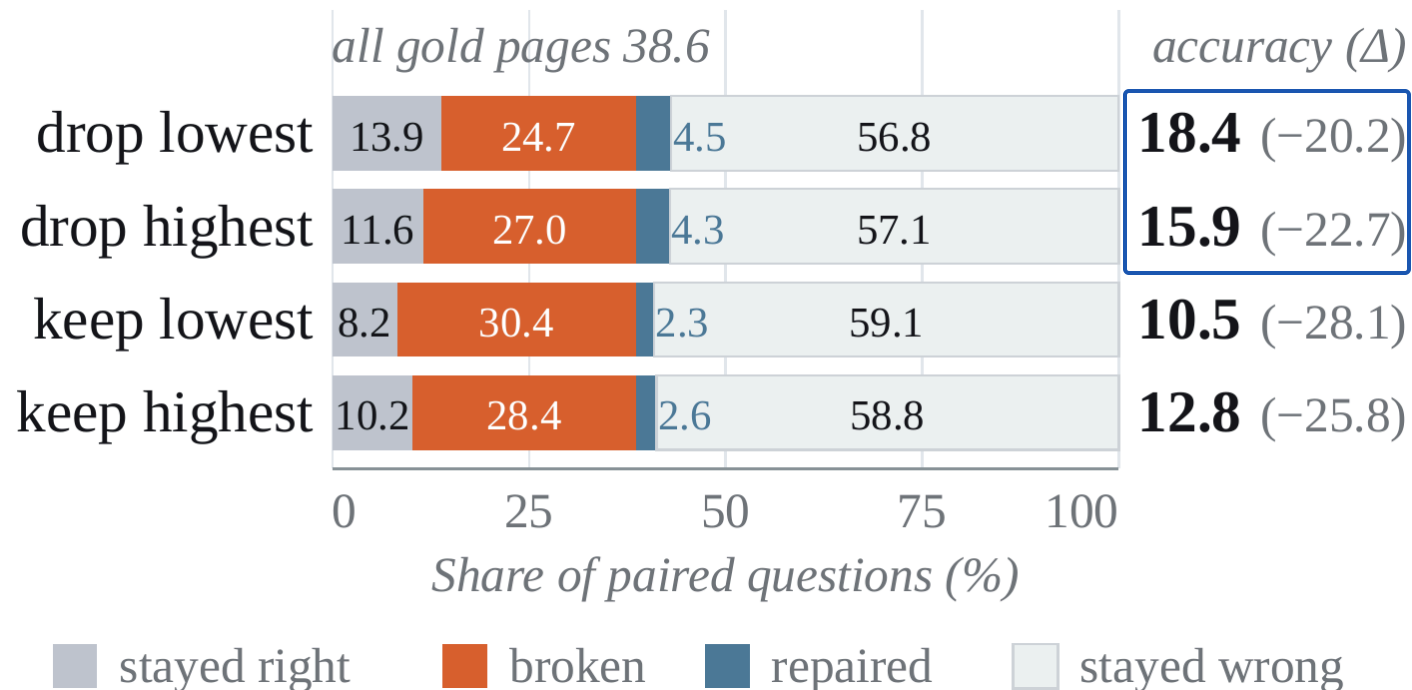
Answer
fixed

REPRESENTATION

Complementary modalities reduce conversion sensitivity

- The parser spread on digital documents shrinks from **15.7 to 4.2 points** once page images are added.
- Higher resolution helps images alone far more than text plus images.

Evidence Coverage

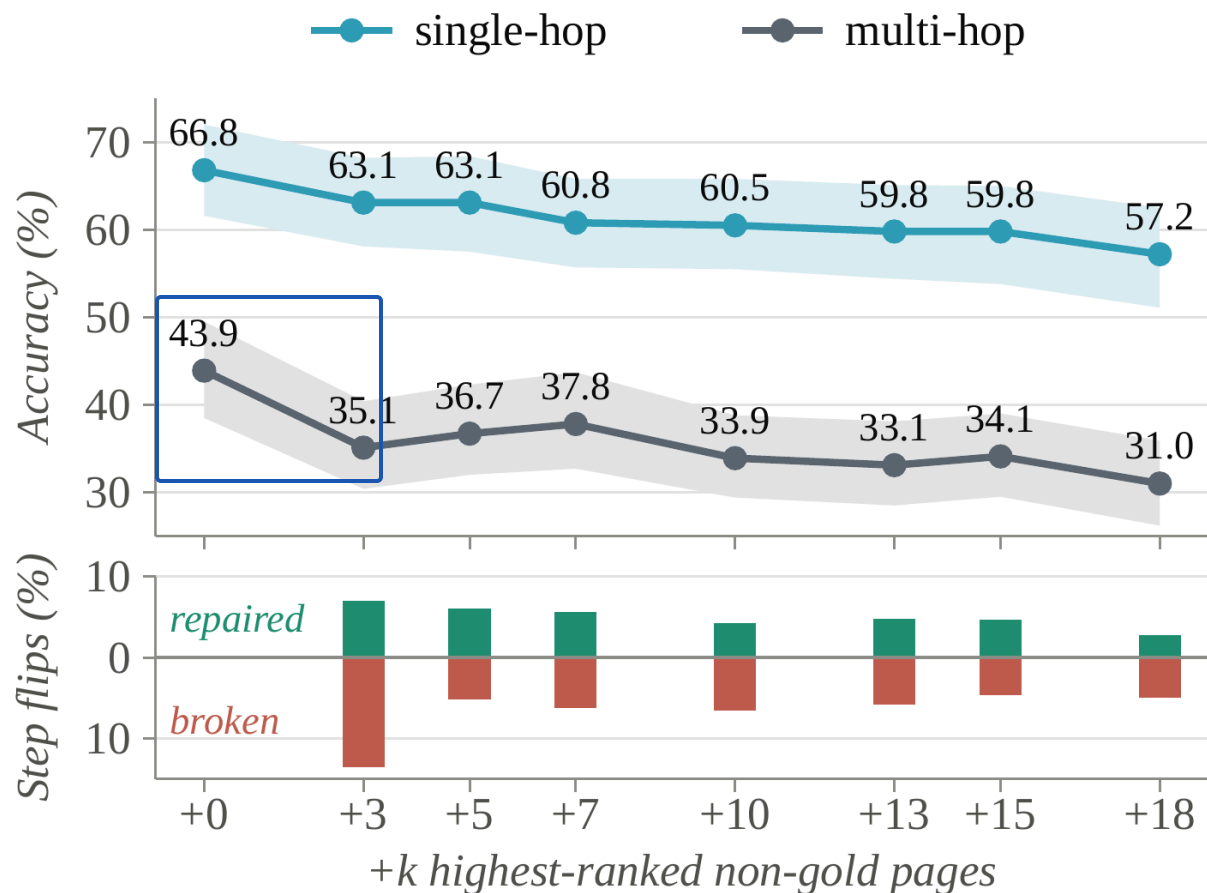


SELECTION

Omission causes large accuracy losses

- Removing **one gold page** takes accuracy from **38.6% to 18.4%** or 15.9%.
- Retrieval rank does not predict which page is safe to lose.

Distractor Exposure



Encode
fixed

Retrieve
gold + k other pages

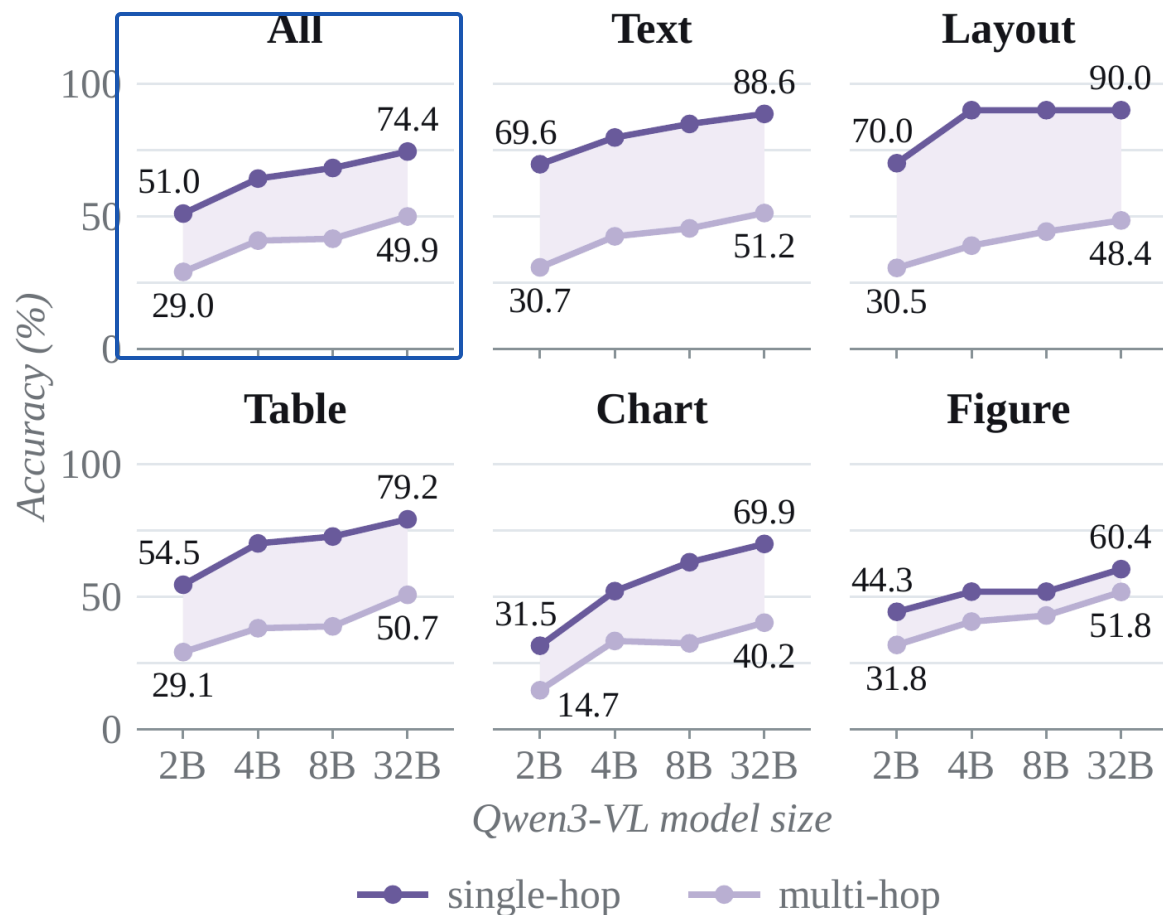
Answer
fixed

SELECTION

Multi-hop accuracy drops sharply with added context

- Three extra pages: multi-hop falls from **43.9%** to **35.1%**. Single-hop declines gradually.
- Every gold page is still there. Added pages both break and repair answers.

Evidence Integration



Encode
fixed

Retrieve
gold pages

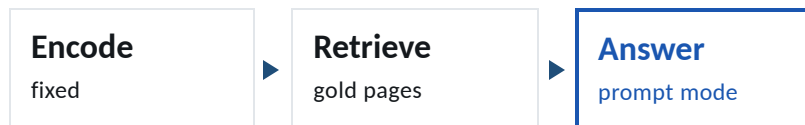
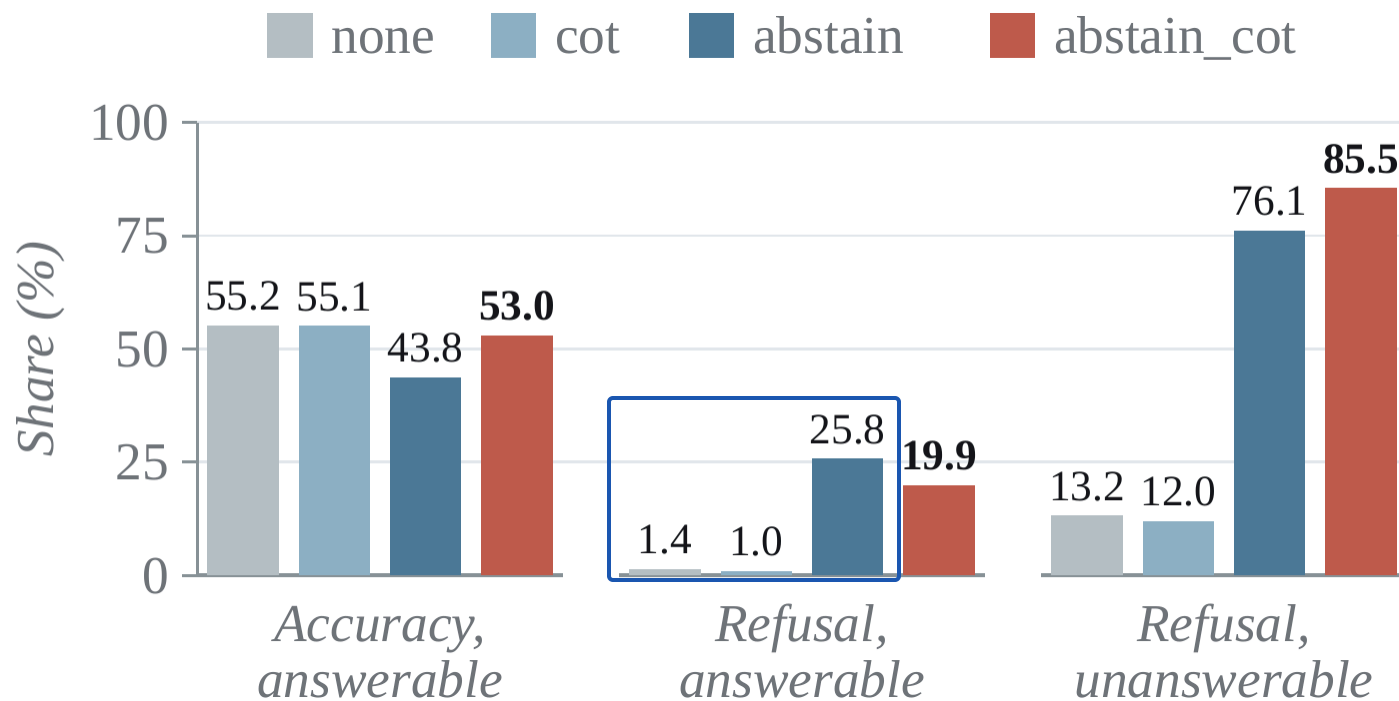
Answer
model size

REASONING

Scale improves multi-hop performance

- From 2B to 32B, multi-hop accuracy rises from **29.0% to 49.9%**.
- Single-hop rises too, so scale helps evidence use in general.

Response Calibration



REASONING

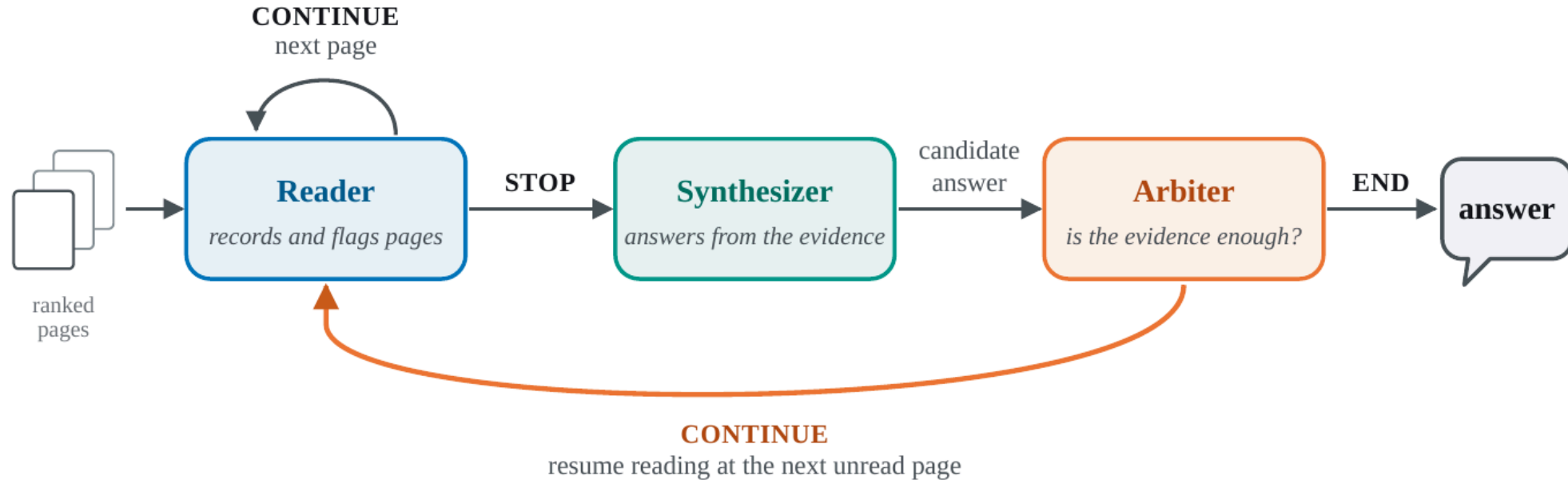
Reasoning before abstaining limits false refusal

- An abstain instruction lifts correct refusal from 13.2% to 76.1%, but false refusal goes from **1.4% to 25.8%**.
- Adding CoT recovers answerable accuracy to 53.0%.

Table of Contents

- Introduction
- Survey of MP-VRDU
- Empirical Attribution Study
- **Proof of Concept: A Simple Method**
- Conclusion

From Findings to a Simple Method



REPRESENTATION

Every page is read as **text and image together**.

SELECTION

Read broadly, then pass on compact records, not full pages.

REASONING

A separate arbiter decides **whether the evidence is enough**.

Simple Method Results

Method	OCR	Trained	MMLongBench-Doc				LongDocURL			
			SIN	MUL	UNA	ACC	UND	REA	LOC	ACC
Gold pages + VLM	✓	×	57.7	37.4	52.5	49.5	69.7	63.6	59.5	65.7
Whole document (via DocTrace)	×	×	46.9	35.3	25.8	38.5	–	–	–	45.1
M3DocRAG (via SLEUTH)	×	×	–	–	48.6	45.9 ^a	60.1	52.4	45.5	54.6
DocAgent (via MARDoc)	✓	×	–	–	–	46.6	–	–	–	–
MDocAgent (via SLEUTH)	✓	×	–	–	49.7	47.8 ^a	57.5	51.5	44.7	53.1
MoLoRAG (via SLEUTH)	×	×	–	–	51.7	48.8 ^a	63.4	52.7	49.3	57.6
SLEUTH	×	×	–	–	67.4	52.8 ^a	65.7	53.0	53.6	60.0
DocTrace	✓	✓	53.2	41.1	70.5	52.9	–	–	–	56.4
Ours (4-bit)	✓	×	60.5	39.2	76.9	56.5	67.2	66.3	47.9	61.3

- **56.5%** on MMLongBench-Doc and **61.3%** on LongDocURL: highest among the Qwen3-VL-8B systems, training-free, at 4-bit.
- Multi-page questions still trail DocTrace, **39.2 against 41.1**.

Conclusion

SURVEY

Organise

Multi-page document understanding is a problem of **evidence management**.

EMPIRICAL STUDY

Locate

Failures sit at **representation, selection, or reasoning**, and each needs a different fix.

PROOF OF CONCEPT

Validate

The findings hold up in a **working, training-free system**.

Thank you Questions?

Inference-Time and Training Strategies

Three evidence-management decisions. Each can be **prompted around a frozen MLLM**, or **trained into it**.

Retrieval and navigation

TRAINING-FREE

Similarity-based or relation-aware retrieval

TRAINED

Modality-alignment, cross-page-fusion, logical-relevance

Reasoning strategies

TRAINING-FREE

Chain-of-Thought, Self-Reflection, Self-Consistency

TRAINED

Trace-imitation, then reward-driven (GRPO)

Agentic methods

TRAINING-FREE

ReAct and tool use, multi-agent orchestration

TRAINED

Trajectory-distillation, reinforcement-learned

Retrieval and Navigation

Retrieval **narrows a long document to compact evidence units** (chunks, pages, or graph nodes) for downstream reasoning.

- **Similarity-based retrieval.** Scores each evidence unit independently against the query and returns the top-ranked subset. Sparse lexical signals such as BM25 are the baseline; dense retrieval uses vision-language late-interaction encoders such as ColPali and ColQwen, often with an LLM reranking stage after.
- **Relation-aware retrieval.** Retrieves along explicit document structure. Tree-based variants follow section hierarchies and outlines, so chunks keep their parent context; graph-based variants traverse typed edges between headings, entities, and cross-references.
- **The trade-off.** Independent scoring handles paraphrase and cross-modal queries, but cannot follow cross-page references or multi-hop dependencies. Relation-aware methods can, at the cost of parser dependence and traversal cost.

Evidence missed or poorly ranked at retrieval is **unrecoverable at generation**. What is retrieved still has to be reasoned over.

Reasoning Strategies

Reasoning strategies act over a **single evidence buffer assembled before the reasoning call**: the full document representation, a retrieved subset, or evidence from an agentic step. None updates model parameters.

- **Sequential reasoning: Chain-of-Thought.** Extends a single reasoning thread. CoT binds evidence from different pages into one chain and decomposes multi-hop questions into single-hop steps.
- **Sequential reasoning: Self-Reflection.** The model evaluates its own draft answer and revises it across rounds, optionally against a verifier or critic.
- **Parallel exploration.** Several reasoning paths over the same buffer. Self-Consistency samples multiple chains and takes the majority answer; sampling-adjudication replaces the vote with an LLM judge. Cost scales with the number of samples. Tree-of-Thoughts has seen little uptake.

A chain can only reason over the evidence already assembled. It **cannot recover what retrieval missed**.

Agentic Methods

Agentic methods put one or more **LLM controllers** in charge: either iterating evidence acquisition under inference-time control, or coordinating across specialised roles.

- **ReAct and tool-augmented reasoning.** A single controller interleaves thought, action, and observation. Actions are document-side tool calls (retrievers, parsers, OCR engines) or structured navigation over a pre-built document representation.
- **Multi-agent orchestration.** Modality-specialised decomposition gives text and visual evidence separate controllers plus a synthesis controller. Role-specialised decomposition splits planning, execution, verification, and synthesis.
- **What the adaptive trajectory buys.** Compute scales with question difficulty rather than document length, the single-step recall ceiling of retrieval-augmented pipelines is lifted, and the action sequence is an inspectable audit trail.

The cost is **controllability**: inference cost is hard to bound, and a controller can commit early to wrong evidence and continue down the wrong trajectory.

Training Strategies

Most systems adopt a backbone **already pretrained on single-page data**, so fine-tuning carries the weight. Training internalises the inference-time evidence-management decisions into model parameters.

Trained retrieval

Modality-alignment aligns visual page embeddings with OCR-derived text. **Cross-page-fusion** adds neighbouring-page context to each page representation. **Logical-relevance** ranks pages by their position in a cross-page reasoning path.

RETRIEVER

Trained reasoning

Trace-imitation supervises the backbone on demonstration chains in a fixed schema. **Reward-driven** training then refines it with GRPO, rewarding answer accuracy, evidence-page recall, and format compliance.

REASONER

Trained agentic methods

Trajectory-distillation clones a stronger teacher's think-action-observation turns through SFT. **Reinforcement-learned** variants optimise the policy directly against trajectory-level rewards.

CONTROLLER

Two techniques recur across all three: **parameter-efficient tuning** (LoRA, adapters) and **multi-stage curricula** that go from single-page to document-scale supervision. All of it depends on the available datasets.

Datasets

Datasets serve three roles, and the multi-page setting makes all three harder: **per-page annotation does not scale** to documents of tens or hundreds of pages, and long PDFs are harder to source and license than single pages.

Pretraining

The exception rather than the norm. Confined to a few page-to-document and MLLM-centric systems that do continued pretraining on an existing backbone, because multi-page corpora are still small.

RARE

Fine-tuning

Has converged on LVLM-synthesised supervision over scraped PDFs, with reinforcement learning increasingly layered on top to shape evidence-grounding and trajectory behaviour.

LVLM-SYNTHESISED

Benchmarking

Three tiers: single-page extractive baselines, mid-length multi-page suites, and long-document multimodal benchmarks that need cross-page integration. The same small pool of PDFs gets reused.

CONTAMINATION RISK

Answer-only accuracy masks retrieval failure and rewards correct answers reached through incorrect reasoning. Newer benchmarks score retrieval and page-localisation as well, but it is not yet standard.

Experimental Setup

One simple pipeline throughout, with **one condition changed at a time** and everything else held fixed.

Encode

Every page becomes text, an image, or both.
Done once per document and cached, so no question changes what a page contains.

PaddleOCR-VL · 200 DPI

Retrieve

▶ A retriever scores every encoded page against the question and returns the top k , restored to document order.

ColQwen3 · page-level

Answer

▶ A multimodal model receives those pages and the question, and answers in a single pass.
Nothing is fed back.

Qwen3-VL-8B · greedy

- **The stages act once, and in order.** That is what lets a failure be attributed rather than guessed at.
- **Gold pages bypass retrieval.** For most tests the annotated evidence pages are supplied directly, so selection is neutralised and whatever remains is representation or reasoning.
- **Answers are scored by an LLM judge**, Gemini-2.5-Flash at temperature zero. It agrees with GPT-4o on 96.0% and with a human annotator on 98.3% of validation subsets.

Datasets and Evaluation

None of these experiments are possible without annotation that says **which pages support each question**. That is what picks the benchmarks.

MMLongBench-Doc

PRIMARY DIAGNOSTIC

135	47.5	847	244
documents	mean pages	answerable	unanswerable

The only multi-page benchmark we know of that annotates, per question, **the evidence pages, the kind of evidence it draws on, and the document's domain**, while also supplying plausible questions the document cannot answer.

Evidence pages make coverage measurable and define the oracle. Evidence source explains representation results by what the question actually needs. The unanswerable pool makes calibration testable rather than inferred.

LongDocURL

SECONDARY REPLICATION

396	85.6	2,325	8
documents	mean pages	questions	unanswerable

Longer documents, used to check whether the findings hold under a **less demanding setting**. Born-digital throughout, almost no unanswerable questions, and many questions answerable from the text even when the annotated evidence is visual.

Its annotated pages are also not a minimal set: several may be redundant or independently sufficient, which we account for when reading its coverage and integration results.

Answers are scored by an LLM judge rather than exact match, since a correct response can differ in units, rounding, or formatting. The judge sees only the question, gold answer, and model answer, never the condition being tested. It agrees with GPT-4o on **96.0%** of a stratified sample and with a human annotator on **98.3%**. We track refusals and wrong answers separately, not just accuracy.

LongDocURL Transfer

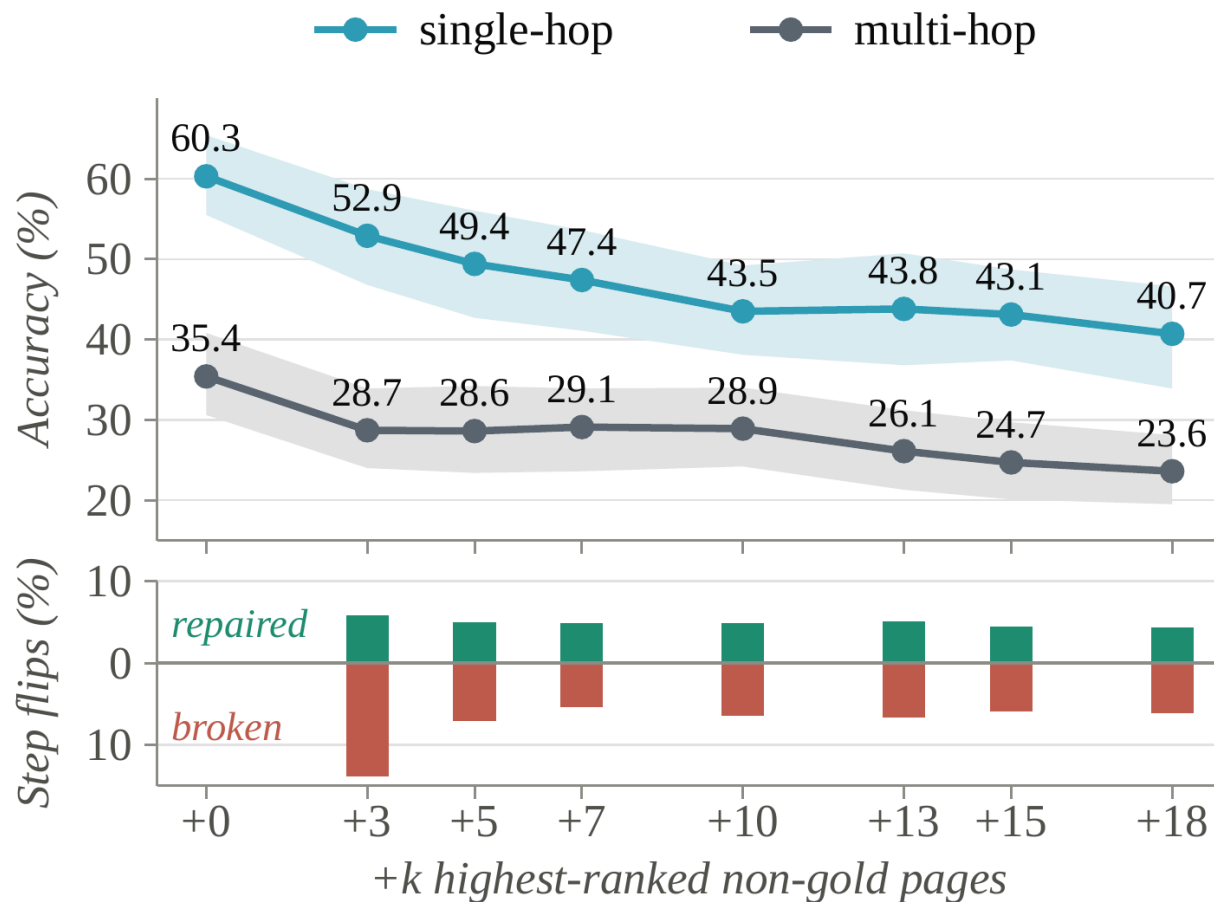
	T	TL	TLV	V
<i>LongDocURL</i>				
Text	75.3	79.5	82.3	74.0
Layout	48.7	52.4	56.3	52.2
Table	61.2	69.5	68.2	57.3
Figure	62.5	65.7	73.4	65.2
All	63.7	68.0	71.9	63.1
<i>MMLongBench-Doc</i>				
Born-digital	40.6	40.1	49.7	41.5
Scanned	4.0	32.8	61.2	58.2

REPRESENTATION

Born-digital documents leave less for vision to add

- Figure questions already reach **65.7%** from parser text. Adding images gives **3.9 points** overall.
- On MMLongBench-Doc, vision adds 28.4 points on scans and 9.6 on born-digital documents.

Distractors on Another Backbone

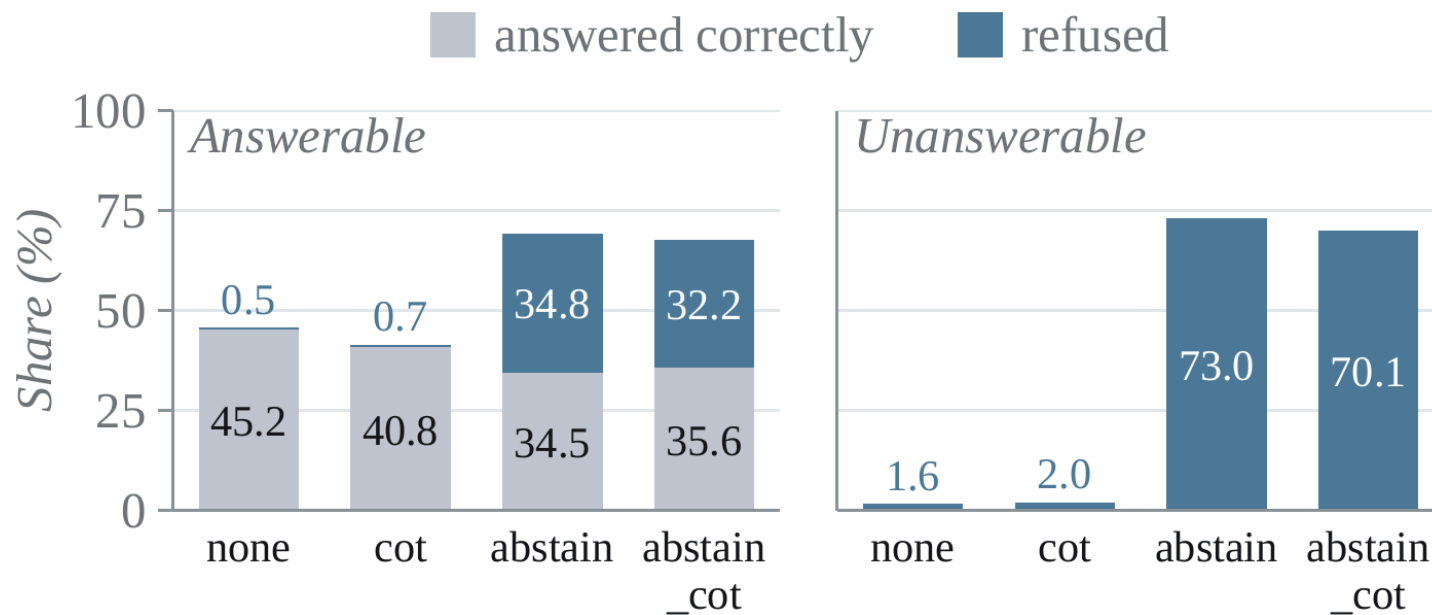


SELECTION

Gemma3-12B shows the same decline

- Single-hop falls from 60.3% to 40.7%, multi-hop from 35.4% to 23.6%.
- Added context still both repairs and breaks answers.

Calibration on Another Backbone



REASONING

Prompt effects depend on the model

- Adding CoT to abstain gives Gemma3-12B **1.1 points** of answerable accuracy, against 9.2 for Qwen3-VL-8B.
- Correct refusal falls 2.9 points for Gemma and rises 9.4 for Qwen.

Representation Cost

	Acc.	Tokens per page			Latency (s)	
		Text	Vis.	Total	Prefill	Decode
T	29.0	499	0	499	1.4	5.7
TL	35.7	874	0	874	1.8	6.6
TV	49.6	500	459	959	—	—
TLV	49.5	874	459	1333	26.3	6.7
V	42.6	20	459	479	29.0	7.1

DEPLOYMENT

Latency follows visual processing,
not token count

- Prefill is **29.0 s** per question with images alone, against 1.4 to 1.8 s for text.
- Embedded text plus images matches parser text plus images: **50.1% against 49.5%**, with no parser.

Reasoner Choice Under Budget

Reasoner	T	TL	TLV	V	GB
<i>Scale (Qwen3-VL, bf16)</i>					
2B	24.7	31.9	38.3	31.9	4.3
4B	32.3	38.6	50.4	42.4	8.9
8B	31.9	38.4	52.4	45.5	17.5
32B	35.5	43.9	59.6	56.7	66.7
<i>Quantized (Qwen3-VL)</i>					
8B, 8-bit	32.5	37.9	53.6	45.7	10.3~
8B, 4-bit	31.8	39.4	51.8	45.5	6.8~
32B, 4-bit	37.0	42.1	59.1	52.3	20.0~
<i>Other families (bf16)</i>					
InternVL3-8B	19.3	25.3	32.6	24.1	15.9
MiniCPM-V-4.5 (8B)	30.7	38.1	60.1	53.3	17.4
GLM-4.6V-Flash (9B)	25.9	32.9	42.7	30.9	20.6
Llama-3.2-11B-Vision	26.5	34.3	33.3	25.3	21.3
Gemma-3-12B	30.5	36.9	48.1	41.6	24.4

DEPLOYMENT

Family and precision matter as well as size

- Qwen3-VL-8B keeps **51.8%** at 4-bit, in about 6.8 GB of weights.
- The 4-bit 32B model reaches **59.1%** in about 20 GB.

From Findings to Design

The survey and the experiments both point at design principles. I put them together in a **deliberately simple system**, to see whether they hold up at the level of a whole pipeline rather than one controlled condition.

REPRESENTATION

Keep text and images together

- ▶ Every page read at the full TLV representation, with a strong parser and high-resolution rendering, since the two channels cover for each other's failures.

SELECTION

Gather broadly, present selectively

- ▶ Missing evidence is unrecoverable and moderate distraction is not, so read widely in retrieval-ranked order, but carry most of it forward as compact records rather than as full pages.

REASONING

Check whether the answer is supported

- ▶ Because the reasoner cannot judge sufficiency on its own, a separate step decides whether to accept the answer or keep reading.

No planner, no tool set, no query reformulation. **The point is to validate the principles, not to introduce a complex architecture.**

Simple Method Cost

	Turns	Tokens			Latency (s)		Peak VRAM (GiB)
		Input text	Input visual	Output	Prefill	Decode	
reader	11.5	5,030	816	697	1.8	46.8	7.2
synthesizer	1.2	9,307	2,018	463	3.8	34.7	8.1
arbiter	1.2	6,806	1,734	464	2.9	33.3	7.9
Per question, mean	13.8	76,669	13,767	9,092	28.5	617.5	8.2
Per question, median	9.0	43,113	9,830	6,078	15.8	385.0	8.0

- **13.8 turns** per question on average, 11.5 of them the reader.
- The cost is time, not memory: **10.8 minutes** mean, 6.7 median, with peak memory at 8.2 GiB.